Survival of the Fittest

How do populations change over time — and what does that have to do with bacteria?

Natural selection is how populations change over generations. Today you'll see it in action with rock pocket mice, then connect it to how bacteria become superbugs.

PART A — Rock Pocket Mouse Data (HHMI)

***Open:** HHMI BioInteractive — "Color Variation Over Time in Rock Pocket Mouse Populations" from your unit page. Use the data to answer the questions below.*

1. Where do dark-furred mice survive better — on light sandy rock or dark lava rock? Explain why using the idea of predation.

2. Look at the population data over time. What happens to the percentage of dark mice in the lava rock area over many generations?

3. Did any individual mouse change its fur color to survive? Explain what actually changed and what didn't.

PART B — VISTA Organizer — Rock Pocket Mice

Fill in each row of VISTA using the rock pocket mouse example.

Letter	Stands for	Rock pocket mouse example
V	Variation	
I	Inheritance	
S	Selection	
T	Time	
A	Adaptation	

PART C — PhET Natural Selection Simulation

Open: *PhET Natural Selection simulation from your unit page. Add wolves as a predator. Run the simulation for at least 10 generations and record what happens.*

What you observe	Write what happened
Starting population (describe fur color mix)	
After 5 generations with wolves	
After 10+ generations with wolves	
What happened to white/light bunnies?	
What would happen if you removed wolves?	

4. What is the "wolf" in the bacteria-antibiotic story? What is the equivalent of "fur color" in bacteria?

PART D — Natural Selection → Superbugs

Natural selection doesn't just happen with animals over thousands of years. It happens with bacteria in days. Here's why that matters:

5. A patient takes an antibiotic but stops after 3 days because they feel better. Of 1,000,000 bacteria, 5 happen to have a mutation that makes them resistant. Using VISTA, explain what happens to those 5 bacteria over the next week.

6. Why do doctors say "always finish your prescription"? Connect your answer to natural selection.

PART E — Helpful, Harmful, or Neither? — Mutations

Mutations are random changes in DNA. They are not "good" or "bad" on their own — it depends on the environment. Sort each mutation below.

Mutation	Environment	Helpful / Harmful / Neither?
Dark fur in a mouse	Dark lava rock	
Dark fur in a mouse	Light sandy desert	

Resistance to penicillin in a bacterium	Hospital with antibiotics	
Resistance to penicillin in a bacterium	Soil with no antibiotics	
Mutation causing no flagellum in a bacterium	Still water with food nearby	

■ **Exit Ticket** In 2–3 sentences, explain how natural selection and antibiotic resistance are connected. Use the words: **variation, mutation, survive, reproduce**.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Simulations: HHMI BioInteractive · PhET Colorado · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html